

Content Focus

Mathematics Foundation

- 3.3.1 Determine whether an accurate answer or an estimation is needed in everyday contexts involving fractions and decimals
- 3.3.2 Choose to add, subtract, multiply or divide fractions and decimals to solve a range of everyday problems involving fractions and decimals
- 3.3.3 Choose between simple decimals and fraction equivalents to solve problems in practical contexts
- 3.3.4 Choose between mental, calculator or spreadsheet to solve problems in practical contexts
- 3.3.5 Mentally solve everyday problems with fractions
- Add and subtract simple fractions mentally by visualising fractional parts and counting
 - Use links between everyday fractions and decimals to assist in mental calculations
- 3.3.6 Use links between everyday fractions and decimals with a calculator when more complex numbers are involved
- 3.3.7 Use a spreadsheet to solve everyday problems involving fractions or decimals
- 3.3.8 Use properties of operations to anticipate the effect of operations on fractions or decimals
- 3.3.9 Use estimation strategies, including rounding, when an accurate answer is not required
- 3.3.10 Interpret decimal remainders from division calculations in relation to the context
- 3.3.11 Determine whether an answer is reasonable using properties of operations, estimation and the context of the problem
- 3.3.12 Communicate solutions using language and symbols consistent with the context

Australian Curriculum Link

- ACMNA077 Investigate equivalent fractions used in contexts
- ACMNA078 Count by quarters, halves and thirds, including with mixed numerals. Locate and represent these fractions on a number line
- ACMNA079 Recognise that the place value system can be extended to tenths and hundredths. Make connections between fractions and decimal notation
- ACMNA083 Use equivalent number sentences involving addition and subtraction to find unknown quantities
- ACMNA099 Use estimation and rounding to check the reasonableness of answers to calculations
- ACMNA121 Use equivalent number sentences involving multiplication and division to find unknown quantities
- ACMNA126 Solve problems involving addition and subtraction of fractions with the same or related denominators
- ACMNA127 Find a simple fraction of a quantity where the result is a whole number, with and without digital technologies
- ACMNA128 Add and subtract decimals, with and without digital technologies, and use estimation without digital technologies, and use estimation and rounding to check the reasonableness of answers
- ACMNA129 Multiply decimals by whole numbers and perform divisions by non-zero whole numbers where the results are terminating decimals, with and without digital technologies
- ACMNA130 Multiply and divide decimals by powers of 10
- ACMNA153 Solve problems involving addition and subtraction of fractions, including those with unrelated denominators
- ACMNA154 Multiply and divide fractions and decimals using efficient written strategies and digital technologies
- ACMNA156 Round decimals to a specified number of decimal places
- ACMNA157 Connect fractions, decimals and percentages and carry out simple conversions

Topic 1

Choosing an Operation to Solve Everyday Problems

Mathematics Discussion

In this section, we focus on which operation: addition, subtraction, multiplication or division to choose, in order to solve everyday problems involving fractions and decimals.

Remember **Addition and Subtraction** problems can be recognised because they usually involve changing an amount by adding more or taking some away, thinking of an amount as a combination of two or more parts, comparing amounts or making two amounts equal.

In addition or subtraction problems, we use Part–Part Whole thinking to decide whether the ‘Whole’ is missing (addition) or a ‘Part’ is missing (subtraction). For example, ‘In a ‘Relay for Life’ charity run, Sue and Kelly ran $5\frac{1}{2}$ km. If Sue ran $3\frac{1}{4}$ km, how far did Kelly run?’

$3\frac{1}{4}$ km is one part of the total run of $5\frac{1}{2}$ km. We can place these numbers in a diagram.

We can write a number sentence that reflects the problem $3\frac{1}{4} + ? = 5\frac{1}{2}$



We could think of this as *what needs to be added to $3\frac{1}{4}$ to get $5\frac{1}{2}$* or, as *a part is missing, $5\frac{1}{2} - 3\frac{1}{4} = ?$*

These are equivalent number sentences.

Multiplication problems can be recognised because they

usually involve repeated equal quantities, arrays, ‘times as many’, rates, or measures (usually area or volume). **Division problems** involve the same situations, and sharing or grouping.

In multiplication and division problems we use arrays and our understanding that:

SIZE OF EACH GROUP $\times$ NUMBER OF GROUPS = TOTAL

- If the TOTAL is missing, we use MULTIPLICATION
- If the TOTAL is present, we use DIVISION

For example, ‘Sue bought 1.3 kg of cashews. The cashews cost \$4.32. How much were the cashews per kilogram?’

The total cost is \$4.32 and the number of groups is 1.3. We can place these numbers in an array diagram. The diagram below shows that we do not know the size of one whole group i.e. the price of one kilogram.

SIZE OF A GROUP (\$/kg)?

TOTAL \$4.32

1.3 kg

We can write a number sentence to reflect how we think about the problem:

$$1.3 \times ? = \$4.32$$

As the TOTAL is present, we can use division. To solve this on a calculator we would enter

$$\$4.32 \div 0.3 = ?$$

Having 'Parts' and 'Wholes' in addition or subtraction problems and 'Sizes of Groups', 'Number of Groups' or 'Totals' in multiplication or division problems that are fractions or decimals can be confusing, but we must remember that we choose the operation in the same way as problems with whole numbers.

Whole Class Activity 1

What key information in a problem involving fractions or decimals helps us choose between addition and subtraction?

The picture in Situation 1 and the table in Situation 2 are examples of where we see fractions and decimals.

SITUATION 1

Sam's Chocolate Mud Cakes at the Biko Cafe



SITUATION 2

The following table shows the penalties that will apply for a first offence of driving with a Blood Alcohol Content (BAC) of greater than or equal to 0.05 and less than 0.08 in Western Australia.

Blood Alcohol Content (g/100 ml)	
Greater than or equal to 0.05 but less than 0.06	3
Greater than or equal to 0.06 but less than 0.07	4
Greater than or equal to 0.07 but less than 0.08	5
Demerit Points	

Robert, a class member, wrote the following problems relating to the diagrams and tables:

1. Sam sold $1\frac{1}{12}$ of a tart. How much tart was left?

2. Tom blew a BAC of 0.062. Katie blew 0.05 less. What was Katie's BAC?

3. Stuart had his BAC tested and 2 hours later another test showed that it was 0.015 less. His BAC now read 0.04. What was Stuart's initial BAC reading?

4. Sam sold $\frac{4}{3}$ of a chocolate mud cake in the morning and $\frac{1}{2}$ in the afternoon. How much cake did Sam sell?

Write Robert's problems in the correct part of the following table. Show your thinking of how to choose the correct operation in the table below.

Discuss your work with your class members.

Fraction Problem	
Addition problem	Number sentence to reflect problem: Part/part whole diagram: 
	Number sentence to find answer (this may be the same number sentence as above):
Decimal Problem	
Addition problem	Number sentence to reflect problem: Part/part whole diagram: 
	Number sentence to find answer (this may be the same number sentence as above):



What are the characteristics of the ADDITION problems?

Write each of your problems in the class poster.

Copy the above table onto a large class poster.

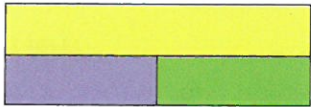
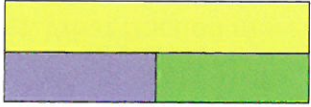
Decimal		
Fraction		
	Addition Problem	Subtraction Problem

Using the chocolate mud cake diagram and the BAC table, write one addition and one subtraction problem in the table below that could be applied to each situation.

Remember, addition and subtraction problems usually involve changing an amount by adding more or taking some away, thinking of an amount as a combination of two or more parts or equalising or comparing amounts.



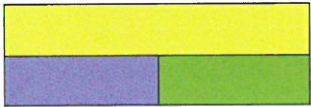
Were there any problems that could be solved using either addition or subtraction? Discuss your thinking.

Subtraction problem	Number sentence to reflect problem:  Part/part whole diagram: Number sentence to find answer (this may be the same number sentence as above):
Subtraction problem	Number sentence to reflect problem:  Part/part whole diagram: Number sentence to find answer (this may be the same number sentence as above):



Practice Exercise 1

1. Show your thinking about the following problems in the table below:

a) PROBLEM: Marika budgets $\frac{2}{7}$ of her pay to savings. Belinda budgets $\frac{1}{10}$ of her pay to savings. What fraction more does Marika budget than Belinda?	Number sentence to reflect problem: Part/part whole diagram:  Number sentence to find answer: Number sentence to enter in calculator:
b) PROBLEM: Marika has a 2 kg lump of cookie dough. She bakes $\frac{3}{4}$ kg of dough. How much dough does she have left?	Number sentence to reflect problem: Part/part whole diagram:  Number sentence to find answer: Number sentence to enter in calculator:

2. Justify which operation, addition or subtraction, best reflects the following problems, by stating whether the whole or a part is missing. The first problem has been completed for you.

a) Su-Lam is 0.19 m shorter than Francesca. If Francesca is 1.8 m tall, how tall is Su-Lam?

PART missing; Subtraction

- b) Paul ran $\frac{1}{4}$ km and then walked $\frac{3}{2}$ km. What fraction of a kilometre has he covered?
- c) Jason drank 0.75 L of water before a 10 km run and 1.3 L after the run. How much water had he drunk altogether?
- d) In a men's javelin competition the winner's two throws totalled 178.4 m. If the first throw was 90.4 m, how long was the second throw?
- e) Two spanners have sizes of $\frac{9}{16}$ inches and $\frac{3}{8}$ inches. What is the difference in these spanner sizes?
- f) Jenny weighed 2.35 kg at birth. In the first week, she gained 0.8 kg. What was her weight at the end of the first week?

3. Write a number sentence to calculate the answers to the following problems:
- a) Belle jumps 3.35 m in Long Jump at the Athletics Carnival and Paul jumps 2.64 m. How much longer does Paul's jump have to be to match Belle's?
 - b) Belle jumps $3\frac{3}{4}$ m and Paul jumps $2\frac{1}{2}$ m. What is their total jumping distance?
 - c) Paul jumps 2.23 m. If he jumps another 0.91 m, he will have jumped the same distance as Belle. How far did Belle jump?
 - d) Belle jumped $\frac{1}{4}$ m less on her second jump than her first. If she jumped $3\frac{3}{8}$ m on her second jump, how much did she jump on her first?
 - e) Belle jumps 3.45 m. If Paul jumps another 0.91 m on his second jump, his jump will be the same as Belle's. What was Paul's original jump?
 - f) Belle and Paul jumped a total of 6 m. If Belle jumped $3\frac{1}{8}$ m, how far did Paul jump?

Whole Class Activity 2

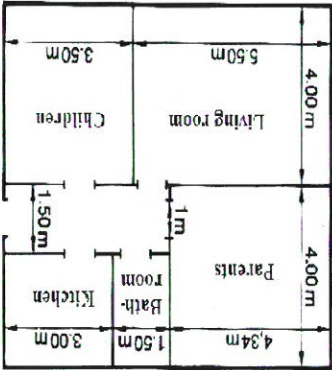
Think: What key information in a problem, involving fractions and decimals, helps us choose between multiplication and division?

The situations below are examples of where we see fractions and decimals in daily life. Roberta, a class member, wrote the following problems:

Situation	Problems
1. Chocolate Chip Cookies 2½ cups of flour 1 tsp baking soda ¾ tsp salt 1 cup of butter ¾ cup of sugar 1 tsp vanilla 2 eggs ¾ cup of chocolate chips Makes 60 cookies.	1. If I made a $\frac{1}{4}$ of the recipe, how much sugar would I need? 2. Jack uses $1\frac{1}{2}$ cups of Chocolate Chips. How many quantities of this recipe is Jack making?

Multiplication Problems	Problem number: Array diagram Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):		Problem number: Array diagram Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):
Division Problems			

Write Roberta's problems in the correct part of the following table. Show your thinking of how to choose the correct operation in the table below. Discuss with your class members.

3. 	3. What is the area of the parent's room in the house floor plan? 4. If the kitchen, in the house floor plan, has an area of 5.25 cm² and its width is 3.00 m, what is its length?
2. 	5. If the chicken costs \$9.60 and Jack bought 0.519 kg, how much was the chicken per kilogram? 6. The chicken costs \$18.49/kg, how much would 2.5 kg cost?



Two students were discussing which column to put question 1 in.
One student said it was multiplication because they thought the *Number sentence to find answer* would be $\frac{3}{4} \times \frac{1}{4} = ?$ The other student said it was division because they thought the *Number sentence to find answer* would be $\frac{3}{4} \div 4 = ?$ Who is correct? Discuss your thinking.

Problem number: Array diagram Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):	Problem number: Array diagram Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):
Problem number: Array diagram Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):	Problem number: Array diagram Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):

Using the recipe, the house floor plan or the meat tray price ticket, write one multiplication and one division problem in the table below.

	Multiplication Problem	Division Problem
Fraction		
Decimal		

Copy the above table onto a large class poster and write each of your problems into it.



What are the characteristics of the MULTIPLICATION problems?



What are the characteristics of the DIVISION problems?

Practice Exercise 2

1. Show your thinking about the following problems in the table below:

a) Josie earns \$450 per week. Ruby earns $\frac{5}{4}$ of this amount. How much does Ruby earn?	Number sentence to reflect problem: Array diagram: Number sentence to find answer (this may be the same as above): Number sentence to enter in calculator:
b) A bus travels 13.24 km to school. There are 11 bus stops evenly placed along this route. How far apart are the bus stops?	Number sentence to reflect problem: Array diagram: Number sentence to find answer (this may be the same as above): Number sentence to enter in calculator:

2. Max made some array sketches that represent multiplication or division problems. Write a problem for each array and complete the required number sentences.

a) Problem: Array diagram Total cost? \$6.25/kg 3.75 kg Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):	b) Problem: Array diagram Total cost \$3.50 \$1.40/L ? L Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):
c) Problem: Array diagram Total cost \$6.40 \$/kg? 0.4 kg Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):	d) Problem: Array diagram Total pizza $6\frac{1}{2}$ Amount per person? 3 people Number sentence to reflect problem: Number sentence to find answer (this may be the same as above):

3. a) Katie is selling oranges at a local farmer's market. She charges \$6.25 per kilogram. Circle the operation she would key into her calculator as she weighed each of these amounts of oranges:

- 4 kg
 - $4\frac{3}{4}$ kg
 - 1.25 kg
 - 0.75 kg
- | | | | |
|--------------------|------------------|--------------------|--------------------|
| $6.25 + 4$ | 6.25×4 | $6.25 \div 4$ | $6.25 - 4$ |
| $6.25 - 4.75$ | $6.25 \div 4.75$ | 6.25×4.75 | $6.25 + 4.75$ |
| $6.25 \div 1.25$ | $6.25 + 1.25$ | $6.25 - 1.25$ | 6.25×1.25 |
| 6.25×0.75 | $6.25 - 0.75$ | $6.25 + 0.75$ | $6.25 \div 0.75$ |

b) Explain your choice of operation in the above problems.

4. a) Katie has 135.6 kg of oranges. Circle the operation she would key into her calculator to work out how many bags she would make up if each bag weighed:

- 4 kg
 - $2\frac{1}{2}$ kg
 - $\frac{3}{4}$ kg
 - 0.625 kg
- | | | | |
|----------------------|------------------|--------------------|---------------------|
| $135.6 + 4$ | 135.6×4 | $135.6 \div 4$ | $135.6 - 4$ |
| $135.6 - 2.5$ | $135.6 \div 2.5$ | 135.6×2.5 | $135.6 + 2.5$ |
| $135.6 \div 0.75$ | $135.6 + 0.75$ | $135.6 - 3 + 4$ | 135.6×0.75 |
| 135.6×0.625 | $135.6 - 0.625$ | $135.6 + 0.625$ | $135.6 \div 0.625$ |

b) Explain your choice of operation in the above problems.

5. Choose between multiplication or division to solve the following problems by stating what is missing – the TOTAL (x), the SIZE OF EACH GROUP or the NUMBER OF GROUPS (÷).

- The first problem has been done for you.
- a) Frank's pace rate is 1.3 paces per metre. How many paces will he take if he walks 100 m?
- b) Indoor netball games take $\frac{3}{4}$ of an hour. How many games could be played in 3 hours?
- c) The scale on a map is 1 cm = 10 km. The distance on the map between towns A and B is $8\frac{1}{4}$ cm. How far apart are the towns?
- d) Each person at a sports event will drink 0.75 litres of juice from a 22 litre container. How many people will get a drink?

TOTAL present; Division

e) A recipe requires $\frac{3}{4}$ cup of butter. How much butter is needed if the recipe is tripled?

f) Belinda cuts an 8.4 m length of rope into 0.6 m pieces. How many pieces will there be?

6. Write a number sentence to calculate the answers to the following problems:

a) An Olympic pool has 8 lanes for competitors and one extra lane on each side. All lanes are $2\frac{1}{2}$ m wide. How wide is an Olympic Pool?

b) The area of one lane of the pool is 125 m^2 . If the width of the lane is 2.5 m, what is the length?

c) A 'warm-up pool' is one quarter the length of a 50 m Olympic Pool. How long is a 'warm-up pool'?

d) Grant Hackett, Olympic Medallist, swims 100 m in 58.3 seconds in long distance events. How long would it take him to swim one 50 m lap of the pool?

e) Denis Pankratov, from Russia, can swim the 50 m Butterfly in 23.68 seconds. Sophie, from the Belmont Swim Club, takes $2\frac{4}{3}$ times longer. How long does Sophie take?

f) Lane lines in pools are wire cables covered with small plastic "floats" that separate each lane. If the floats are 1.8 m apart, how many sections of cable are there between the floats in one lane line in a 50 m pool?

Practice Exercise 3

1. Circle the addition and subtraction problems in blue. Write whether a 'PART' is missing; SUBTRACTION' or a 'WHOLE' is missing; ADDITION', next to the problem.

Circle the multiplication and division problems in red. Write whether the 'TOTAL' is missing; MULTIPLICATION' or 'TOTAL' is present; DIVISION', next to the problem.

a) A bike weighs 9.3 kg. How much would 4 of the same bikes weigh?

b) The average standing jump of a 10 year old boy is 145.5 cm. Billy can jump 2.7 cm longer. How far can Billy jump?

c) A bottle of soft drink is $\frac{4}{3}$ litre. How many litres would 19 of these bottles contain?

d) 84.3 kg of sugar is poured into bags which hold 1.5 kg each. How many bags can be filled?

e) Jan has a bag of lollies weighing $\frac{7}{8}$ kg. Keryn's bag weighs $\frac{1}{2}$ kg. How much more does Jan's bag weigh than Keryn's?

f) Bec pours $3\frac{1}{2}$ litres of tea into $\frac{1}{4}$ litre cups. How many cups can she pour?

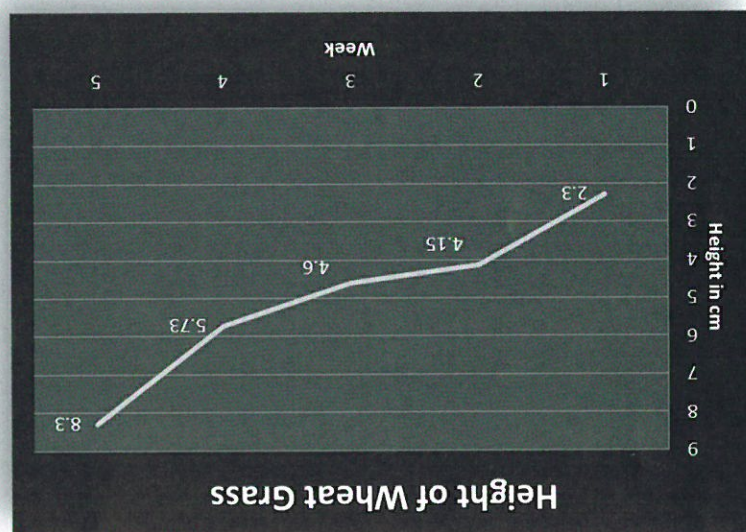
2. Match the word problems with one of the following number sentences:

a) A. $\frac{3}{4} + \frac{1}{2}$

B. $\frac{4}{3} - \frac{1}{2}$

C. $\frac{4}{3} \times \frac{1}{2}$

D. $\frac{3}{4} \div \frac{1}{2}$



3. The following graph shows the 'Height of Wheat Grass' as it grows over a 5 week period. Write a number sentence next to each question that could be used to calculate the answer.

- b)
- Daniel paid \$32.35 for petrol. If petrol cost \$1.40 per litre, how many litres did Daniel buy?
 - Daniel filled his car with 32.35 litres of petrol. He used 1.4 litres of petrol driving to the shops. How much petrol did Daniel have left in his car?
 - Daniel bought 32.35 litres of petrol costing \$1.40 per litre. How much did this cost?
 - Daniel bought 32.35 litres of petrol for his car and 1.4 litres for his lawnmower. How much petrol did Daniel buy?
 - Daniel bought a 32.35 L large container full of petrol and poured it into smaller 1.4 L containers. How many smaller containers were there?
- A. $32.35 + 1.4$ B. $32.35 - 1.4$ C. 32.35×1.4 D. $32.35 \div 1.4$
- $\frac{3}{4}$ of a pizza was leftover from a party. Stephanie ate $\frac{1}{2}$ of that amount. What fraction of a pizza did Stephanie eat?
 - Peter ate $\frac{1}{2}$ a pizza. Rebecca ate $\frac{4}{3}$ of a pizza. How much pizza did they eat altogether?
 - $\frac{3}{4}$ of a pizza was leftover from a party. Ronald ate some and a half was left. How much did Ronald eat?
 - $\frac{1}{2}$ of a pizza is considered, by a pizza company, as one serve. How many serves would $\frac{4}{3}$ of a pizza be?
 - Mary cut $\frac{3}{4}$ of a pizza in half. What is the size of the halved piece?

$\frac{3}{4} + 1\frac{1}{2}$	T	4.73×0.7	W
$1\frac{1}{2} - \frac{4}{3}$	T	10.5×1.5	A
$8.99 \div 1.4$	S	$4.73 \div 0.7$	Y
$10.5 \div 1.5$	I	$1\frac{1}{2} \div \frac{4}{3}$	P
$\frac{4}{3} - 2$	V	$\frac{4}{3} \times 2$	C
$\frac{4}{3} \times \frac{8}{3}$	R	1.4×8.99	O
$\frac{4}{3} - \frac{8}{3}$	M	$10.5 + 1.5$	A
$\frac{4}{3} \div 2$	E		L

What did the hamburger name his daughter?

2. In the puzzle below, circle the correct number sentence used to calculate the answers to each of the problems above. Delete all incorrect number sentences and their corresponding letters. You will be left with the answer to the following joke:

- f) Kristen separates 10.5 kg of kiwifruit into 1.5 kg bags. How many bags will she make up?
- e) Jen buys $\frac{4}{3}$ kg of peaches and shares them evenly with John. How much do they each get?
- d) Bronte buys $\frac{4}{3}$ kg of oranges. Peter buys $\frac{8}{3}$ kg oranges. How much do they buy altogether?
- c) Peter buys 1.4 kg of bananas at \$8.99/kg. How much does this cost?
- b) Corey buys $\frac{4}{3}$ kg of apricots. Stacey buys $1\frac{2}{3}$ kg more. How much does Stacey buy?
- a) Sandra buys 0.7 kg of apples. It costs \$4.73. How much are the apples per kilogram?

Try to do the thinking mentally. Use jottings and diagrams to help if needed.

- Between addition/subtraction and multiplication/division
- For addition/subtraction problems whether the 'Part' is missing (subtraction) or the 'Whole' is missing (addition)
- For multiplication/division problems whether the 'Total' is present (division) or the 'Total' is missing (multiplication)

1. For the following problems decide:

Reflection on Learning

- a) How much did the wheat grass grow between weeks 4 and 5?
- b) How many times taller was the wheat grass in the third week than the first?
- c) How many times smaller was the wheat grass in the second week than the fifth?
- d) The wheat grass grew 1.5 times its size from week 5 to week 6. How tall was it in week 6?
- e) The wheat grass grew 1.13 cm from weeks 3 to 4. True or false? Justify your answer.

OLNA Practice Questions

1. Mary weighs 1.25 kg more than Michelle. If Michelle weighs 68.35 kg, how much does Mary weigh? Which operation should be used to find the answer to this problem?

- A. ADDITION B. SUBTRACTION C. MULTIPLICATION D. DIVISION

2. Harry buys 2.5 kg of plums and it costs \$17.46. The number sentence Harry would key into his calculator to find the price of plums per kilogram would be:

- A. $17.46 + 2.5$ B. $17.46 \div 2.5$ C. $17.46 - 2.5$ D. 17.46×2.5

3. Harry buys 0.3 kg oranges at \$6.25/kg. The number sentence Harry would key into his calculator to find the amount he needs to pay would be:

- A. $6.25 + 0.3$ B. $6.25 \div 0.3$ C. $6.25 - 0.3$ D. 6.25×0.3